

Ayphen HMS — Complete Application Flow Reference

Version: 2026-05-08 | Stack: NestJS + Prisma + PostgreSQL (port 6666) · React + TypeScript + Vite + Tailwind CSS (port 5555)

Application Summary

- **Ayphen HMS** is a cloud-based, multi-tenant Hospital Management System (SaaS) designed to serve multiple hospitals and clinics from a single deployment.
- Every record is isolated by `tenantId`, ensuring complete data separation between organisations.
- The system covers the full hospital workflow — from patient registration and appointment booking through clinical care, pharmacy, lab, billing, and discharge.
- Built on a **NestJS + Prisma + PostgreSQL** backend (port 6666) and a **React + TypeScript + Vite + Tailwind CSS** frontend (port 5555).
- Supports **11 distinct user roles**: Platform Owner, Platform Admin, Admin, Doctor, Nurse, Receptionist, Pharmacy, Lab, Billing, OT, and Patient.
- **Four separate login paths** exist: staff login, doctor login (multi-org support), patient self-service login, and platform admin login.
- Doctors can be affiliated with multiple organisations and choose their active org at login.
- Patients register and log in through a dedicated **Patient Portal** with a clean layout (no staff sidebar), allowing them to book appointments, view prescriptions, lab reports, billing, and vitals.
- The appointment booking system shows real-time available slots (30-min intervals, 08:00–17:00) by doctor and date, with Sundays blocked.
- **JWT-based authentication** with access + refresh tokens, optional TOTP-based MFA, rate limiting, and FCM push notification support.
- The frontend has **107 pages** and the backend has **72 modules** covering over 350 API endpoints.
- A global **Command Palette** (Cmd+K) enables cross-module search across patients, appointments, invoices, prescriptions, admissions, lab orders, and queue entries.
- Pharmacy uses **FEFO (First-Expiry-First-Out)** batch depletion for inventory management.
- The **Medication Administration Record (MAR)** enforces a 5-Rights verification checklist (Patient, Drug, Dose, Route, Time) before any dose is recorded.
- Clinical modules include: Emergency & Triage (ESI scoring), ICU/NICU, Operating Theatre scheduling, Dialysis, Physiotherapy, Wound Care, Palliative Care, Telemedicine, and Blood Bank.
- Operational modules cover: CSSD sterilization tracking, Waste Management manifests, Work Orders, Duty Roster with shift swap approvals, Housekeeping, Linen, and Asset Management.
- Compliance and record-keeping modules include: Audit Log, Birth/Death Records, MLC Register, Infection Control, Antimicrobial Stewardship, Clinical Pathways, and Quality Management.

- HR modules include Staff Attendance, Payroll, Duty Roster, and Shift Handover.
- The **Reports** module generates summaries across 8 categories: Lab, Pharmacy, Appointments, Inventory, Billing, Admissions, Staff, and Quality — with CSV export and print support.
- Platform Admins manage tenant onboarding, feature flags, and cross-tenant analytics from a dedicated platform dashboard.

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1. System Architecture

Overview

Ayphen HMS is a **multi-tenant SaaS hospital management system**. Every record is scoped by a `tenantId` (Organisation). A single deployment serves unlimited hospitals/clinics.

Browser (React SPA) | Vite Dev Server :5555 | REST + WebSocket | NestJS API :6666 | Prisma ORM | PostgreSQL (Supabase) | JWT Auth (access + refresh tokens) | Throttle Guard (rate limiting) | WebSocket Gateway (ws-gateway) | Firebase Cloud Messaging (push notifications)

Key Architectural Decisions

Concern	Solution
----- -----	
Multi-tenancy	Every table has <code>tenantId</code> ; all queries filter by it
Auth identity	TenantUser (staff), DoctorAccount (doctors), PatientAccount (patients), PlatformUser (super admin)
Patient records	<code>PatientAccount</code> (login) ≠ <code>Patient</code> (clinical). Linked by email via <code>resolvePatientRecord()</code>

Sequential IDs	<code>generateSequentialId()</code> utility used across all modules (e.g., PAT-0001 , RX-0042)
Role isolation	Frontend <code>RoleRoute</code> + backend <code>@Roles()</code> guard on every sensitive endpoint
Code splitting	<code>React.lazy()</code> + <code>Suspense</code> wraps every page component in <code>App.tsx</code>

2. Authentication & Roles

2.1 Login Flows

There are **four separate login paths**:

Staff Login (`POST /auth/login`)

- Used by: ADMIN, NURSE, RECEPTION, PHARMACY, LAB, BILLING, OT, etc.
- Flow:

1. Submit email + password → server validates `TenantUser` , returns `accessToken` + `refreshToken` 2. If MFA enabled → `POST /auth/mfa/verify` with TOTP code 3. Token stored in `localStorage` via `auth.ts` 4. Redirect to `/app/dashboard`

Doctor Login (`POST /auth/doctor/login`)

- Doctors may be affiliated with multiple organisations
- Flow:

1. Submit credentials → returns `accessToken` with `type: 'doctor'` 2. If single affiliation → automatic org selection 3. If multiple → show org selector → `POST /auth/doctor/select-org` with `affiliationId` 4. New token issued with `tenantId` scoped to selected org 5. Redirect to `/app/doctor/consultations`

Patient Login (`POST /auth/patient/login`)

- Flow:

1. Patient registers via `POST /auth/patient/register` (creates `PatientAccount`) 2. Login → token with `type: 'patient'` 3. If linked to multiple orgs → `POST /auth/patient/select-org` with `tenantId` + `locationId` 4. Redirect to `/app/patient/portal`

Platform Login (`POST /auth/platform/login`)

- Used by: PLATFORM_OWNER, PLATFORM_ADMIN
- No `tenantId` in token — sees all organisations
- Redirect to `/platform/dashboard`

2.2 Token Management

POST /auth/refresh → exchange refreshToken for new accessToken GET /auth/me → current user profile PATCH /auth/me → update name/phone PUT /auth/me/password → change password POST /auth/forgot-password → send reset email POST /auth/reset-password → consume reset token POST /auth/device-token → register FCM push token DELETE /auth/device-token → unregister FCM push token

2.3 MFA (TOTP)

POST /auth/mfa/setup → generate QR code secret POST /auth/mfa/activate → confirm code to enable POST /auth/mfa/verify → verify code on login

2.4 User Roles & Permissions

Role	Scope	Main Pages
----- ----- -----		
PLATFORM_OWNER	All tenants	Platform dashboard, tenants, users, features
PLATFORM_ADMIN	All tenants	Same as PLATFORM_OWNER
ADMIN	Own tenant	All staff pages + admin settings
DOCTOR	Own tenant (scoped)	Consultations, prescriptions, lab orders, OT scheduling
NURSE	Own tenant	MAR, vitals, ward rounds, wound care, shift handover
RECEPTION	Own tenant	Appointments, queue, admissions, visitors
PHARMACY	Own tenant	Dispense, inventory, purchase indents
LAB	Own tenant	Lab orders, results, QC, calibration
BILLING	Own tenant	Invoices, insurance, health packages
OT	Own tenant	OT scheduling, CSSD
PATIENT	Own org (PatientAccount)	Portal, appointments, records, prescriptions, billing

3. Platform Admin Flows

3.1 Tenant Management

GET /tenants → list all organisations POST /tenants → create new organisation (name, subdomain, plan) GET /tenants/:id → org detail + stats PATCH /tenants/:id → update org settings DELETE /tenants/:id → deactivate org GET /tenants/:id/stats → usage metrics

Flow: Platform admin logs in → /platform/dashboard shows org count, user count, active subscriptions → Navigate to /platform/tenants → Create/edit orgs → Assign features.

3.2 Feature Flags

GET /features → all feature flags POST /features → create flag PATCH /features/:id → toggle on/off DELETE /features/:id → remove flag GET /features/tenant/:tenantId → flags for org POST /features/tenant/:tenantId → assign flag to org

3.3 Platform Users

GET /platform/users → list platform users POST /platform/users → create platform admin PATCH /platform/users/:id → update DELETE /platform/users/:id → remove

3.4 Platform Reports & Analytics

GET /platform/analytics → cross-tenant usage stats GET /platform/audit-log → platform-level audit trail

4. Staff Portal — Core Clinical Flows

4.1 Patient Registration & Admission

Registration

- Reception opens **Patients** page → GET /patients (paginated, searchable)
- Click "New Patient" → form: name, DOB, gender, phone, address, blood group, emergency contact
- Submit → POST /patients → auto-generates patientId (e.g., PAT-0001)
- Patient card created with tenantId + locationId

GET /patients → list (search by name/phone/ID) POST /patients → create GET /patients/:id → full profile PATCH /patients/:id → update demographics DELETE /patients/:id → soft-delete GET /patients/:id/history → clinical history timeline

Admission

- From patient profile or Admissions page → POST /admissions
- Select ward, bed, admitting doctor, diagnosis
- Generates admissionNumber (e.g., ADM-0042)
- Bed status set to OCCUPIED

GET /admissions → active admissions list POST /admissions → admit patient GET /admissions/:id → admission detail PATCH /admissions/:id → update (ward transfer, diagnosis change) POST /admissions/:id/discharge → discharge with summary

4.2 Appointment Booking (Staff Side)

GET /appointments → list with filters (date, doctor, status) POST /appointments → create booking GET /appointments/:id → detail PATCH /appointments/:id → reschedule / update status PATCH /appointments/:id/cancel →

cancel with reason GET /appointments/slots → available slots for doctor+date `

Slot Logic: GET /appointments/slots?doctorId=&date= checks existing bookings and returns 30-min slots 08:00–17:00, marking each available: true/false .

4.3 Doctor Consultation Flow

- Doctor logs in → sees today's appointment queue
- Opens **Consultation** page for a patient
- Records: chief complaint, history, examination findings, diagnosis (ICD-10)
- **Inline Lab Orders:** selects tests directly in consultation panel
- **Prescriptions:** adds drugs with dosage/frequency/duration
- Saves → POST /consultations

` GET /consultations → list POST /consultations → create (with vitals, notes, ICD codes) GET /consultations/:id → full consultation PATCH /consultations/:id → update POST /consultations/:id/lab-orders → attach lab orders `

4.4 Prescription Flow

- Doctor creates prescription during consultation or standalone
- POST /prescriptions with doctorId , patientId , items array
- Generates rxNumber (e.g., RX-0007)
- Pharmacy sees pending prescriptions → dispense

` GET /prescriptions → list (filter by status, patient, doctor) POST /prescriptions → create GET /prescriptions/:id → detail with items PATCH /prescriptions/:id → update POST /prescriptions/:id/dispense → mark as dispensed (pharmacy) GET /prescriptions/:id/items → line items `

4.5 Medication Administration (MAR)

- Nurse opens **MAR** page → sees scheduled medications by patient/ward
- Overdue doses highlighted in amber/red
- Click "Administer" → **5-Rights Checklist** popup:
 - Right Patient, Right Drug, Right Dose, Right Route, Right Time
- Confirm all 5 → mark administered
- Click "Withhold" → modal for reason → PATCH /medication-admin/:id/withhold

` GET /medication-admin → schedule list (filter by ward/date) POST /medication-admin → create schedule entry PATCH /medication-admin/:id → update administration record PATCH /medication-admin/:id/withhold → withhold with reason `

4.6 Lab Workflow

- Doctor orders tests (inline in consultation or standalone)
- **Lab** page shows pending orders
- Lab tech enters results: value, flag (HIGH/LOW/CRITICAL/NORMAL), reference range
- Results auto-flagged → doctor notified

GET /lab/orders → pending/completed orders POST /lab/orders → create order GET /lab/orders/:id → order with results PATCH /lab/orders/:id → update POST /lab/orders/:id/results → enter results GET /lab/results → all results GET /lab/panels → test panels POST /lab/panels → create panel GET /lab/qc/runs → QC run log POST /lab/qc/runs → log QC run (auto PASS/WARN/FAIL) GET /lab/qc/calibrations → calibration log POST /lab/qc/calibrations → log calibration

4.7 Radiology Workflow

GET /radiology/orders → imaging orders POST /radiology/orders → create order PATCH /radiology/orders/:id → update POST /radiology/orders/:id/report → attach report/image GET /radiology/results → completed studies

4.8 Pharmacy Workflow

- Pharmacist sees pending prescriptions → GET /prescriptions?status=PENDING
- Verifies drug, dose, patient allergy (barcode scan via ZXing supported)
- Dispenses → POST /prescriptions/:id/dispense
- Stock decremented via **FEFO** (First-Expiry-First-Out) batch depletion
- Low-stock alert if quantity falls below threshold

GET /pharmacy/inventory → drug stock levels GET /pharmacy/low-stock → items below threshold POST /pharmacy/dispense → dispense medication GET /pharmacy/batches → inventory batches (FEFO) POST /pharmacy/batches → add batch GET /pharmacy/reports → dispensing reports

4.9 Billing Flow

- Billing staff opens invoice list → GET /billing/invoices
- Create invoice → select patient, add line items (consultation, drugs, lab, bed charges)
- Apply insurance or health package discount
- Mark paid → PATCH /billing/invoices/:id with status: 'PAID'

GET /billing/invoices → invoice list POST /billing/invoices → create invoice GET /billing/invoices/:id → detail with items PATCH /billing/invoices/:id → update / mark paid POST /billing/invoices/:id/email → email invoice PDF GET /billing/summary → revenue summary GET /billing/pending → unpaid invoices

4.10 Insurance

GET /insurance/claims → claims list POST /insurance/claims → submit claim PATCH /insurance/claims/:id → update claim status GET /insurance/providers → insurer list POST /insurance/providers → add insurer GET /insurance/preauth → pre-authorization requests POST /insurance/preauth → request pre-auth

4.11 Emergency & Triage

- Patient arrives at emergency → Triage nurse assesses
- POST /triage with vitals + chief complaint → auto-assigns ESI level (1–5)
- Assigned to queue → doctor picks up

GET /triage → triage queue POST /triage → create triage record PATCH /triage/:id → update triage / assign bed GET /emergency → emergency cases POST /emergency → register emergency case PATCH /emergency/:id → update

4.12 Operating Theatre (OT)

GET /ot/schedules → OT schedule POST /ot/schedules → book OT slot PATCH /ot/schedules/:id → update (surgeon, anaesthetist, room) POST /ot/schedules/:id/start → mark surgery start POST /ot/schedules/:id/complete → mark surgery complete GET /ot/rooms → OT rooms

4.13 ICU & NICU

GET /icu → ICU patient list POST /icu → admit to ICU PATCH /icu/:id → update ICU record (vitals, ventilator, drips) POST /icu/:id/transfer → transfer out of ICU GET /nicu → NICU patient list POST /nicu → admit neonate PATCH /nicu/:id → update NICU record

4.14 Ward Management

GET /wards → ward list POST /wards → create ward GET /wards/:id/beds → beds in ward with occupancy POST /wards/:id/beds → add beds PATCH /wards/:id/beds/:bedId → update bed status

4.15 Discharge Summary

GET /discharge-summary → list POST /discharge-summary → create (with diagnosis, treatment, follow-up) GET /discharge-summary/:id → full summary PATCH /discharge-summary/:id → update POST /discharge-summary/:id/print → generate PDF

5. Patient Portal

5.1 Layout

Patients use a **dedicated layout** (`PatientLayout`) — no staff sidebar. A clean top navigation bar with 7 items:

- Dashboard (Portal)
- Appointments
- Medical Records
- Prescriptions
- Lab Reports
- Billing
- Vitals

All routes live under `/app/patient/*` and require `PATIENT` role.

5.2 Patient Registration (Self-Service)

POST /auth/patient/register → create PatientAccount POST /auth/patient/login → login POST /auth/patient/select-org → link to hospital (tenantId + locationId)

5.3 Booking Appointments


```
GET /auth/patient/me/appointments → list with doctorName enrichment PATCH /appointments/:id/cancel → cancel booking
```

```
GET /auth/patient/me/appointments → appointment history POST /auth/patient/me/appointments → book appointment
GET /auth/patient/me/prescriptions → prescription history GET /auth/patient/me/lab-reports → lab results GET
/auth/patient/me/billing → invoices and payment status GET /auth/patient/me/vitals → latest vitals GET
/auth/patient/me/profile → demographic profile`
```

Administrative & Platform

Module	Key Endpoints	Frontend Page
----- ----- -----		
Auth	/auth/login, /auth/doctor/login, /auth/patient/login, /auth/me	LoginPage, StaffRegisterPage
Tenants	/tenants CRUD, /tenants/:id/stats	/platform/tenants
Platform	/platform/users, /platform/analytics	/platform/dashboard
Features	/features CRUD, /features/tenant/:id	/platform/features
Users	/users CRUD (staff users)	AdminUsersPage
Roles	/roles CRUD	AdminRolesPage
Departments	/departments CRUD	AdminDepartmentsPage
Locations	/locations CRUD	AdminLocationsPage
Audit	/audit list, filters by module/user/action	AuditPage
Notifications	/notifications list, mark read, push via FCM	NotificationsPage

Reports	/reports/lab , /reports/pharmacy , /reports/appointments , /reports/inventory	ReportsPage (8 tabs)
Search	/search?q= (patients, appointments, invoices, prescriptions, admissions, lab orders, queue)	CommandPalette (Cmd+K)
Uploads	/uploads (file storage)	Used across modules

Clinical Modules

Module	Key Endpoints	Frontend Page
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|-----|-----|-----|

Patients	/patients CRUD, /patients/:id/history	PatientsPage
Appointments	/appointments CRUD, /appointments/slots	AppointmentsPage
Consultations	/consultations CRUD, inline lab orders	ConsultationPage
Prescriptions	/prescriptions CRUD, dispense	PrescriptionsPage, PatientPrescriptionsPage
Medication Admin (MAR)	/medication-admin CRUD, /medication-admin/:id/withhold	MARPage (Nurse)
Lab	/lab/orders , /lab/results , /lab/qc/runs , /lab/qc/calibrations	LabOrdersPage, LabResultsPage, LabQCPage
Radiology	/radiology/orders , /radiology/results	RadiologyPage
Vitals	/vitals CRUD	VitalsPage, PatientVitalsPage
Admissions	/admissions CRUD, discharge	AdmissionsPage
Discharge Summary	/discharge-summary CRUD	DischargeSummaryPage
Wards	/wards CRUD, /wards/:id/beds	WardsPage

Specialty Clinical

Module	Key Endpoints	Frontend Page
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|-----|-----|-----|

Emergency	/emergency CRUD	EmergencyPage
Triage	/triage CRUD, ESI scoring	TriagePage
ICU	/icu CRUD, transfer	ICUPage
NICU	/nicu CRUD	NICUPage

OT	/ot/schedules CRUD, start/complete	OTSchedulePage
Dialysis	/dialysis CRUD	DialysisPage
Physiotherapy	/physiotherapy CRUD	PhysiotherapyPage
Wound Care	/wound-care CRUD	WoundCarePage
Palliative Care	/palliative-care CRUD, home visits	PalliativeCarePage
Home Care	/home-care CRUD	HomeCarePage
Telemedicine	/telemedicine CRUD, room URL	TelemedicinePage
Ambulance	/ambulance CRUD	AmbulancePage
Blood Bank	/blood-bank CRUD	BloodBankPage
Diet	/diet CRUD, meal plans	DietPage

Administrative Clinical

Module	Key Endpoints	Frontend Page
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|-----|-----|-----|

Billing	/billing/invoices CRUD, email	BillingPage
Insurance	/insurance/claims , /insurance/preauth	InsurancePage
Health Packages	/health-packages CRUD	HealthPackagesPage
Queue	/queue CRUD, token management	QueuePage
MLC Register	/mlc-register CRUD	MLCRegisterPage
Consent	/consent CRUD	ConsentPage
Certificates	/certificates CRUD	CertificatesPage
Discharge Summary	/discharge-summary CRUD	DischargeSummaryPage
Referral	/referral CRUD	ReferralPage
Feedback	/feedback CRUD	FeedbackPage
Grievance	/grievance CRUD	GrievancePage
Visitors	/visitors CRUD	VisitorsPage

Operations & Support

Module	Key Endpoints	Frontend Page
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|-----|-----|-----|

Inventory	<code>/inventory</code> CRUD, batches, FEFO	InventoryPage (6 tabs)
Pharmacy	<code>/pharmacy</code> CRUD, dispense, low-stock	PharmacyPage
Purchase Indent	<code>/purchase-indent</code> CRUD, approve/reject	PurchaseIndentPage
Central Store	<code>/central-store</code> CRUD	CentralStorePage
Vendor	<code>/vendor</code> CRUD	VendorPage
Asset Management	<code>/asset-management</code> CRUD	AssetManagementPage
CSSD	<code>/cssd/batches</code> CRUD, issue/return	CssdPage
Linen	<code>/linen</code> CRUD	LinenPage
Housekeeping	<code>/housekeeping</code> CRUD	HousekeepingPage
Waste Management	<code>/waste-management</code> CRUD, manifest pipeline	WasteManagementPage
Work Orders	<code>/work-orders</code> CRUD, status workflow	WorkOrdersPage

HR & Compliance

Module	Key Endpoints	Frontend Page
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|-----|-----|-----|

Staff Attendance	<code>/staff-attendance</code> CRUD	StaffAttendancePage
Duty Roster	<code>/duty-roster</code> CRUD, swap request/approve	DutyRosterPage
Payroll	<code>/payroll</code> CRUD	PayrollPage
Shift Handover	<code>/shift-handover</code> CRUD	ShiftHandoverPage
Doctor Registry	<code>/doctor-registry</code> CRUD	DoctorRegistryPage
Doctor Affiliations	<code>/doctor-affiliations</code> CRUD	DoctorAffiliationsPage
MRD	<code>/mrd</code> CRUD	MRDPage
Audit	<code>/audit</code> log	AuditPage
Birth/Death	<code>/vital-records/births</code> , <code>/vital-records/deaths</code> CRUD + edit modal	BirthDeathPage
Infection Control	<code>/infection-control</code> CRUD	InfectionControlPage
Quality	<code>/quality</code> CRUD	QualityPage
Antimicrobial Stewardship	<code>/antimicrobial-stewardship</code> CRUD	AntimicrobialPage
Clinical Pathways	<code>/clinical-pathways</code> CRUD	ClinicalPathwaysPage

Mortuary	/mortuary CRUD	MortuaryPage
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7. Data Model Summary

Core Entities

PlatformUser id, email, passwordHash, role (PLATFORM_OWNER/PLATFORM_ADMIN) Tenant id, name, subdomain, plan, active Location id, tenantId, name, type (BRANCH/CLINIC/HOSPITAL) TenantUser id, tenantId, locationId, email, passwordHash, role, firstName, lastName DoctorAccount id, email, passwordHash (cross-tenant identity) DoctorAffiliation id, doctorAccountId, tenantId, locationId, specialization PatientAccount id, email, passwordHash (self-service login) Patient id, tenantId, locationId, patientNumber, firstName, lastName, DOB, gender

Clinical Records

Appointment id, tenantId, patientId, doctorId, date, time, status, type Consultation id, tenantId, patientId, doctorId, appointmentId, diagnosis, notes, icdCodes Prescription id, tenantId, patientId, doctorId, rxNumber, status PrescriptionItem id, prescriptionId, drugName, dosage, frequency, duration, qty MedicationAdministration id, tenantId, patientId, prescriptionId, scheduledTime, dosage, status Admission id, tenantId, patientId, admissionNumber, wardId, bedId, admittingDoctorId Discharge id, tenantId, admissionId, dischargeDate, summary, followUp LabOrder id, tenantId, patientId, doctorId, orderNumber, orderedAt, status LabResult id, labOrderId, testName, value, flag, referenceRange Vitals id, tenantId, patientId, bp, pulse, temperature, spo2, weight, height, recordedAt

Pharmacy & Inventory

InventoryItem id, tenantId, name, category, unit, quantity, reorderLevel InventoryBatch id, inventoryItemId, batchNumber, expiryDate, quantity (FEFO depletion) PharmacyItem id, tenantId, drugName, quantity, unitPrice, reorderLevel

Billing

Invoice id, tenantId, patientId, invoiceNumber, netTotal, status, paidAt InvoiceItem id, invoiceId, description, quantity, unitPrice, total InsuranceClaim id, tenantId, patientId, invoiceId, claimNumber, status, amount

Operations

WorkOrder id, tenantId, title, description, status, assignedTo, priority, dueDate CssdBatch id, tenantId, itemName, batchNumber, sterilizedAt, condition (STERILE/ISSUED/QUARANTINE) WasteCollection id, tenantId, category, weight, manifestNumber, status DutyRoster id, tenantId, userId, date, shiftStart, shiftEnd, swapStatus

8. API Endpoint Map

Authentication

Method	Path	Auth	Description
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|-----|-----|-----|-----|

POST	/auth/login	Public	Staff login
POST	/auth/platform/login	Public	Platform admin login
POST	/auth/doctor/login	Public	Doctor login
POST	/auth/doctor/select-org	JWT	Select doctor's org
POST	/auth/patient/register	Public	Patient self-register
POST	/auth/patient/login	Public	Patient login
POST	/auth/patient/select-org	JWT	Link patient to org
POST	/auth/mfa/setup	JWT	Init MFA
POST	/auth/mfa/activate	JWT	Enable MFA
POST	/auth/mfa/verify	JWT	Verify MFA code
POST	/auth/forgot-password	Public	Send reset email
POST	/auth/reset-password	Public	Reset with token
POST	/auth/refresh	Public	Refresh access token
GET	/auth/me	JWT	Current user profile
PATCH	/auth/me	JWT	Update profile
PUT	/auth/me/password	JWT	Change password
GET	/auth/patient/me/appointments	JWT	Patient's appointments
POST	/auth/patient/me/appointments	JWT	Book appointment
GET	/auth/patient/me/prescriptions	JWT	Patient's prescriptions
GET	/auth/patient/me/lab-reports	JWT	Patient's lab reports
GET	/auth/patient/me/billing	JWT	Patient's invoices
GET	/auth/patient/me/vitals	JWT	Patient's vitals
GET	/auth/patient/me/profile	JWT	Patient's profile
POST	/auth/device-token	JWT	Register FCM token
DELETE	/auth/device-token	JWT	Unregister FCM token

Patients

Method	Path	Description
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|-----|-----|-----|

GET	/patients	List (search, paginate)
POST	/patients	Register new patient
GET	/patients/:id	Patient profile
PATCH	/patients/:id	Update demographics
DELETE	/patients/:id	Soft delete
GET	/patients/:id/history	Clinical timeline

Appointments

Method	Path	Description
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|-----|-----|-----|

GET	/appointments	List with filters
POST	/appointments	Create booking
GET	/appointments/:id	Detail
PATCH	/appointments/:id	Update/reschedule
PATCH	/appointments/:id/cancel	Cancel
GET	/appointments/slots	Available slots

Clinical

Method	Path	Description
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|-----|-----|-----|

GET/POST	/consultations	List/create
GET/PATCH	/consultations/:id	Detail/update
GET/POST	/prescriptions	List/create
PATCH	/prescriptions/:id	Update
POST	/prescriptions/:id/dispense	Pharmacy dispense
GET/POST	/medication-admin	MAR list/create
PATCH	/medication-admin/:id	Administer
PATCH	/medication-admin/:id/withhold	Withhold with reason

GET/POST	/lab/orders	Orders list/create
POST	/lab/orders/:id/results	Enter results
GET/POST	/lab/qc/runs	QC runs
GET/POST	/lab/qc/calibrations	Calibrations
GET/POST	/vitals	Vitals list/create
GET/POST	/admissions	Admissions
POST	/admissions/:id/discharge	Discharge

Billing & Insurance

Method	Path	Description
GET/POST	/billing/invoices	Invoices
PATCH	/billing/invoices/:id	Update/pay
POST	/billing/invoices/:id/email	Email invoice
GET	/billing/summary	Revenue summary
GET/POST	/insurance/claims	Claims
PATCH	/insurance/claims/:id	Update claim
GET/POST	/insurance/preauth	Pre-authorizations

Operations

Method	Path	Description
GET/POST	/inventory	Stock list/create
GET/POST	/inventory/batches	Batch tracking (FEFO)
GET/POST	/work-orders	Work orders
PATCH	/work-orders/:id	Update status
GET/POST	/cssd/batches	CSSD sterilization batches
PATCH	/cssd/batches/:id/issue	Issue to department
PATCH	/cssd/batches/:id/return	Return from department

GET/POST	/waste-management	Waste manifest
GET/POST	/duty-roster	Shift schedule
POST	/duty-roster/:id/swap	Request shift swap
PATCH	/duty-roster/:id/approve-swap	Approve swap
GET/POST	/vital-records/births	Birth records
GET/POST	/vital-records/deaths	Death records
PATCH	/vital-records/births/:id	Edit birth record
PATCH	/vital-records/deaths/:id	Edit death record

Reports

Method	Path	Description
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GET	/reports/lab	Lab report summary
GET	/reports/pharmacy	Pharmacy dispensing report
GET	/reports/appointments	Appointment statistics
GET	/reports/inventory	Inventory usage report
GET	/reports/billing	Revenue report
GET	/reports/admissions	Admission statistics
GET	/reports/staff	Staff activity report
GET	/reports/quality	Quality metrics report

Search

Method	Path	Description
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GET	/search	Global search across 7 entity types: patients, appointments, invoices, prescriptions, admissions, lab orders, queue

Appendix: Application URL Map

URL Pattern	Component	Role(s)
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	LandingPage	Public
/login	LoginPage	Public
/doctor/login	LoginPage (doctor mode)	Public
/patient/login	LoginPage (patient mode)	Public
/platform/login	LoginPage (platform mode)	Public
/app/dashboard	AdminDashboard	ADMIN
/app/patients	PatientsPage	ADMIN, RECEPTION, NURSE
/app/appointments	AppointmentsPage	RECEPTION, ADMIN
/app/queue	QueuePage	RECEPTION
/app/admissions	AdmissionsPage	ADMIN, RECEPTION
/app/billing	BillingPage	BILLING, ADMIN
/app/insurance	InsurancePage	BILLING, ADMIN
/app/pharmacy	PharmacyPage	PHARMACY
/app/inventory	InventoryPage	PHARMACY, ADMIN
/app/lab/orders	LabOrdersPage	LAB
/app/lab/results	LabResultsPage	LAB, DOCTOR
/app/lab/qc	LabQCPage	LAB
/app/radiology	RadiologyPage	LAB, ADMIN
/app/doctor/consultations	ConsultationPage	DOCTOR
/app/doctor/prescriptions	PrescriptionsPage	DOCTOR
/app/doctor/telemedicine	TelemedicinePage	DOCTOR
/app/nurse/mar	MARPage	NURSE
/app/nurse/vitals	VitalsPage	NURSE
/app/nurse/wound-care	WoundCarePage	NURSE
/app/nurse/shift-handover	ShiftHandoverPage	NURSE
/app/ot	OTSchedulePage	OT, ADMIN
/app/icu	ICUPage	NURSE, DOCTOR
/app/nicu	NICUPage	NURSE, DOCTOR
/app/emergency	EmergencyPage	NURSE, DOCTOR, RECEPTION

/app/triage	TriagePage	NURSE
/app/cssd	CssdPage	OT
/app/duty-roster	DutyRosterPage	ADMIN
/app/staff-attendance	StaffAttendancePage	ADMIN
/app/payroll	PayrollPage	ADMIN
/app/work-orders	WorkOrdersPage	ADMIN
/app/waste-management	WasteManagementPage	ADMIN
/app/birth-death	BirthDeathPage	ADMIN, RECEPTION
/app/audit	AuditPage	ADMIN
/app/reports	ReportsPage	ADMIN, BILLING
/platform/dashboard	PlatformDashboard	PLATFORM_OWNER, PLATFORM_ADMIN
/platform/tenants	TenantsPage	PLATFORM_OWNER, PLATFORM_ADMIN
/platform/features	FeaturesPage	PLATFORM_OWNER, PLATFORM_ADMIN
/app/patient/portal	PatientPortalPage	PATIENT
/app/patient/appointments	PatientAppointmentsPage	PATIENT
/app/patient/records	PatientMedicalRecordsPage	PATIENT
/app/patient/prescriptions	PatientPrescriptionsPage	PATIENT
/app/patient/lab	PatientLabReportsPage	PATIENT
/app/patient/billing	PatientBillingPage	PATIENT
/app/patient/vitals	PatientVitalsPage	PATIENT

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